SLASSCOM National Ingenuity Awards 2026 Submission Master Pack for E2ETune++

Project Title: E2ETune++: Domain-Adapted End-to-End Database Knob Tuning Using LLMs and Transfer LearningProject Type: University Final Year Project**Prepared For:** SLASSCOM National Ingenuity Awards 2026**Recommended Categories:**

1. **Best Innovative Product - University**
2. **Best Tech for Good Software Innovation**

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1. Executive Summary

E2ETune++ is an AI-driven database tuning system designed to make advanced database performance optimization more accessible, affordable, and practical across different database engines and hardware environments.

Modern digital services depend on databases, yet databases are notoriously difficult to tune well. Performance depends on many internal parameters, or **"knobs,"** which influence memory usage, disk I/O, concurrency, query execution, and overall responsiveness. In practice, identifying the right configuration usually requires deep DBA expertise, time-consuming experimentation, and repeated system testing. This is especially difficult for small organizations, public institutions, startups, and educational environments that do not have access to expensive database specialists.

Based on the analyzed project materials, **E2ETune++ combines multiple advanced technologies into a single research-driven system**:

- **LLM-based database tuning**
- **Transfer learning**
- **Workload clustering**
- **Cost modeling**
- **Cross-domain adaptation**
- **Cross-database generalization across PostgreSQL and MySQL**
- **Cross-hardware adaptation across multiple server environments**

The project extends **E2ETune**, a recently proposed end-to-end database knob tuning framework that uses a fine-tuned language model to recommend database configurations from workload features. E2ETune++ goes further by addressing a major real-world limitation: **database optimization tools often work only for one database engine and one hardware setting**. That makes them less useful in real-world environments, where organizations use diverse databases and limited computing resources.

E2ETune++ tackles this gap through a layered innovation strategy:

1. **Representative workload selection** using SQL embeddings and clustering, so the team does not need to run every possible workload.
2. **Cost models** that predict database performance without requiring full live experimentation every time.
3. **Transfer learning** from a large PostgreSQL dataset to a much smaller MySQL dataset.
4. **Domain adaptation techniques**, including regression-based transfer learning and **Domain-Adversarial Neural Networks (DANN)**, to improve generalization across database and hardware domains.
5. **LLM fine-tuning strategies**, including supervised fine-tuning and reinforcement-style optimization concepts, to move toward practical end-to-end recommendation.

This project is award-worthy for five reasons:

- **It addresses a real and costly technical problem** faced by almost every data-driven organization.
- **It is technically deep**, combining AI, systems engineering, databases, and transfer learning.
- **It has measurable results**, including strong PostgreSQL cost prediction and clear gains in MySQL adaptation using transfer learning.
- **It has social relevance**, because it can reduce reliance on scarce expert DBAs and make optimization more accessible to resource-constrained organizations.
- **It is forward-looking**, with clear potential to evolve into a practical product for SMEs, educational institutions, public sector systems, and cloud database support tools.

In short, **E2ETune++ is not just a research exercise**. It is a meaningful step toward **democratizing expert-level database optimization using AI**.

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2. Category Selection Strategy

Recommended Category Positioning

Category	Fit Level	Why It Fits	Caution
Best Innovative Product - University	**Very Strong**	Specifically designed for undergraduate student innovations; values originality, feasibility, impact, and project quality	Must emphasize student-led innovation, technical rigor, and future potential
Best Tech for Good Software Innovation	**Strong if framed well**	Can fit if positioned around reducing waste, improving accessibility, and democratizing database optimization	Must clearly articulate social/environmental good, not just technical novelty

Why Best Innovative Product - University Is the Natural Fit

This is the most straightforward and strategically strong category because:

- The project is a **final year undergraduate product/project**
- It has clear **research depth and technical feasibility**
- It demonstrates **original extension of existing state-of-the-art work**
- It has **real-world problem relevance**
- It showcases **teamwork, experimentation, and innovation**

For judges in this category, the story is naturally compelling:
“A university team took a cutting-edge research problem and built a sophisticated, practical AI system that improves on existing approaches.”

Why Best Tech for Good Software Innovation Is Still Valid

E2ETune++ can also fit here **if the narrative is intentionally shaped around social benefit**.

The strongest Tech for Good framing:

- **Democratizes expert-level database optimization**
- **Helps organizations that cannot afford specialist DBAs**
- **Supports SMEs, universities, government institutions, and non-profits**
- **Reduces computational waste from poor database configurations**
- **Improves infrastructure efficiency**
- **Helps bridge the gap between advanced AI infrastructure and under-resourced environments**

This is especially strong in a developing-country context, where the cost of infrastructure inefficiency and lack of specialist expertise has a bigger downstream effect.

Final Recommendation

If budget allows, apply to both:

1. **Primary recommendation: Best Innovative Product - University**
2. **Secondary recommendation: Best Tech for Good Software Innovation**

This gives you:

- A **high-fit category** for judges who value academic innovation
- A **broader-impact category** for judges who value social good and national relevance

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3. Project Understanding from the Provided Materials

3.1 What the Project Is

E2ETune++ is a university research and engineering project that extends **E2ETune**, an LLM-based end-to-end database knob tuning approach. The original E2ETune concept treats knob tuning like a sequence-generation problem: given a representation of a workload, the model predicts a promising database configuration.

E2ETune++ builds on that idea and focuses on a harder, more realistic challenge:

> **How can an AI-based tuning system generalize across different database engines and different hardware environments without requiring massive new data collection every time?**

That is the central contribution.

3.2 Core Technical Components Identified from the Materials

A. LLM-Based End-to-End Tuning

The project uses an **LLM fine-tuning approach inspired by Mistral-7B-style models**, following the E2ETune line of work. The idea is to move from:

- manual DBA tuning, or
- expensive iterative search,

toward **direct prediction of promising knob configurations from workload features**.

B. Workload Embeddings and Clustering

The team does not attempt to run all possible workloads. Instead, it uses:

- **SQL normalization**
- **Embeddings** using **gemini-embedding-001**
- **HDBSCAN clustering**

This is used to select representative workloads from benchmark sets. The purpose is to make training data collection more efficient while keeping the dataset representative.

C. Bayesian Optimization / HEBO for Data Generation

The materials indicate the use of **HEBO**, a Bayesian optimization technique, to search configuration space and generate high-value performance data. This helps create labeled examples that can later be used to train cost models and fine-tune the LLM.

D. Cost Models for Performance Prediction

A major part of the project involves training **cost models** that can predict workload latency/performance. These models reduce dependence on repeated live execution and make tuning workflows more scalable.

E. Cross-Database Transfer Learning

A major challenge is that the project has:

- a **large PostgreSQL dataset**
- a **much smaller MySQL dataset**

So the team investigates whether knowledge learned from PostgreSQL can help on MySQL.

F. Domain Adaptation

The project also explores:

- **Regression-based transfer learning**
- **Residual adapters**
- **Joint weighted training**
- **DANN (Domain-Adversarial Neural Networks)**
- **Semi-supervised domain adaptation**

This is important because database engine differences and hardware differences create domain shift.

3.3 Benchmarks Mentioned

The project materials mention the following benchmarks:

- **JOB**
- **SSB**
- **SSB_FLAT**
- **TPC-H**
- **TPC-DS**
- **SMALLBANK**
- **TPCC**
- **TWITTER**
- **WIKIPEDIA**
- **YCSB**

These span both **OLAP** and **OLTP** style workloads, which strengthens the project's credibility and breadth.

3.4 Hardware Context Mentioned in the Materials

Two server settings were explicitly described:

Server	CPU	RAM	Storage
Server 1	Intel Xeon E3-1275v5, 4 cores / 8 threads, 3.60 GHz	64 GB DDR4	512 GB NVMe SSD
Server 2	Intel Core i7-7700, 4 cores / 8 threads, 3.60 GHz	32 GB DDR4	512 GB SATA SSD

The project also discusses practical infrastructure constraints such as:

- shared cloud environments
- non-identical hardware compared to prior papers

- the cost of recollecting data
- limited GPU resources for fine-tuning

These are realistic engineering constraints and should be presented honestly.

3.5 Dataset and Modeling Context

PostgreSQL / MySQL Regression Study

From the analyzed materials:

- **PostgreSQL workloads:** 217 workloads, **20,758 samples**
- **MySQL workloads:** 46 workloads, **1,314 samples**

Input features include:

- query plan embeddings
- knob configurations
- workload-derived features
- hardware specs
- database type flag

This is strong evidence that the team tackled a non-trivial transfer learning problem with real imbalance between source and target domains.

3.6 Regression-Based Transfer Learning Explored

The project explored:

- **LightGBM**
- **RandomForest**
- **HistGradientBoosting**

For MySQL adaptation, it evaluated:

- **MySQL-only baseline**
- **Residual Adapter**
- **Joint Weighted Training**

These experiments are important because they show the team did not jump to a single fancy method. They tested multiple alternatives and identified what worked best.

3.7 DANN / Domain-Adversarial Learning Explored

The project also ran a separate DANN-based study for cost modeling.

Key technical details extracted:

- **10,190 samples**
- **4 domains**
- **79 features**
- **447 encoded dimensions**
- **548,165 parameters**

It explored:

- **Source-only DANN**
- **SSDA with 10% labeled target data**

- **Supervised baseline without adversarial learning**
- **Global normalization**
- **Per-engine normalization**
- Multiple seeds for robustness

This shows a serious attempt at **cross-domain generalization**, even where results were mixed.

3.8 What the Project Is Really Solving

The deeper project problem is not only “database tuning.”

It is:

> **How to build an AI-driven configuration system that still works when the database engine changes, when hardware changes, and when available training data is uneven across environments.**

That makes the project more important than a simple tuning demo.

3.9 Realistic Limitations and Honest Framing

A strong submission should also acknowledge that this is an advanced undergraduate project and an ongoing research effort. Based on the materials, realistic limitations include:

- Not all components may be fully productized yet
- Some results are stronger than others
- DANN results appear mixed and sensitive to normalization and random seed choice
- Hardware differences from prior published work may affect direct comparison
- Social impact is currently strongest as a **potential-enabling impact**, unless validated with real deployments
- Some productization claims require evidence from the team before use

This honesty will increase credibility.

Best honest framing:

- **Strong research-backed prototype**
- **Practical system direction**
- **High-value AI systems innovation**
- **Clear path to real-world deployment**

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4. Winning Strategy for SLASSCOM

4.1 How Judges Are Likely to Think

SLASSCOM judges are unlikely to reward a project **just because it is technically complex**. They will likely look for a combination of:

1. **A meaningful problem**
2. **A clear and understandable solution**
3. **Evidence that it works**
4. **Potential for practical relevance**
5. **A convincing reason it matters beyond the lab**

Judges will ask, implicitly:

- Why should I care?
- Who benefits?
- What makes this new?
- Is this real, or just experimental?
- Can this grow into something useful?
- Why is this more significant than a standard university research project?

Your submission must answer all of those.

4.2 What to Emphasize

A. Problem Significance

Explain that bad database tuning is not a niche problem. It affects:

- application speed
- server efficiency
- cloud costs
- system scalability
- ability of small organizations to operate modern digital services

B. Real-World Applicability

Frame E2ETune++ as a tool that can support:

- SMEs
- SaaS startups
- universities
- research labs
- public institutions
- low-resource tech teams

C. Technical Depth

Use the technical sophistication as proof of seriousness, not as the main story.

D. Measurable Results

Results make the project credible. The strong regression and transfer learning results should be brought forward clearly.

E. Social Value

This is especially essential for **Best Tech for Good**. The social-good angle is not charity. It is:

- making expert optimization accessible
- reducing infrastructure inefficiency
- helping under-resourced organizations do more with less

4.3 Positioning Advice

Do **not** frame this as:

- “We tried some ML models for a database paper”
- “We reproduced E2ETune”
- “We built a final year research experiment”

Instead, frame it as:

> **E2ETune++ is an AI system that reduces the expertise barrier in database optimization by using transfer learning and LLM-based tuning to adapt across database engines and hardware settings.**

That sounds practical, relevant, and scalable.

4.4 How to Frame This So It Sounds Like a Winner

Use messaging like:

- **“We are tackling a high-impact infrastructure problem that most organizations face but few can solve well.”**
- **“Our project brings expert-level database optimization closer to organizations that cannot afford specialist DBA expertise.”**
- **“We are not only predicting database performance; we are making optimization more transferable across real-world environments.”**
- **“This work bridges AI research and practical systems engineering.”**
- **“The project reduces the cost and complexity of tuning across heterogeneous database environments.”**
- **“Our contribution is especially relevant to resource-constrained environments, where efficiency matters most.”**

Avoid messaging like:

- “Our model got a better RMSE”
- “We used DANN and LightGBM”
- “This is our FYP and we worked hard”

Those points matter, but they do not win awards by themselves.

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5. Full Submission Draft for “Best Tech for Good Software Innovation”

> **Important:** Replace all bracketed placeholders before submission.

5.1 Detailed Write-Up on the Innovative Product/Project

(Target: up to 1000 words)

Draft Answer

E2ETune++: Domain-Adapted End-to-End Database Knob Tuning Using LLMs and Transfer Learning

E2ETune++ is an AI-driven software innovation designed to make advanced database performance optimization more accessible, affordable, and adaptable across different computing environments. The project addresses a major challenge in modern digital infrastructure: database systems are critical to almost every organization, but tuning them properly requires deep expertise, repeated experimentation, and significant time and computing cost.

Database systems such as PostgreSQL and MySQL contain many internal parameters, often called “knobs,” which control memory allocation, I/O behavior, caching, query execution, concurrency, and related performance factors. These settings have a major impact on system responsiveness, throughput, and resource usage. However, identifying good configurations is difficult. In many real-world settings, organizations either depend on experienced database administrators or continue running on default or poorly tuned settings. This creates inefficiency, higher operational cost, slower systems, and avoidable resource waste.

E2ETune++ was developed to address this problem by extending the idea of end-to-end AI-based database tuning. The project builds on E2ETune, a recent research direction that uses a fine-tuned language model to translate workload characteristics directly into promising database configurations. Our contribution is to push this idea toward a more realistic and socially meaningful setting: rather than assuming one database engine and one hardware environment, E2ETune++ explores how tuning knowledge can be adapted across different database systems and different server setups.

The project combines several components into one pipeline. First, we use workload representation and clustering to reduce the cost of collecting training data. SQL workloads are normalized, embedded using a language embedding model, and clustered using HDBSCAN so that representative workloads can be selected rather than exhaustively executing every possible scenario. This improves efficiency while preserving diversity across benchmark types.

Second, we use Bayesian optimization through HEBO to generate high-value training data from benchmark executions. This data is used to build cost models that predict workload latency or performance from workload features, knob settings, hardware information, and database type. These cost models are important because they reduce the need for expensive repeated live database execution during later optimization stages.

Third, E2ETune++ investigates transfer learning from data-rich environments to data-scarce environments. In our project materials, PostgreSQL has a substantially larger dataset than MySQL. Rather than treating MySQL as an entirely separate problem, we explore how knowledge from PostgreSQL can help improve prediction and tuning for MySQL. We evaluate multiple regression-based transfer approaches, including a MySQL-only baseline, a residual adapter strategy, and joint weighted training. The results show that cross-database transfer can materially improve performance prediction quality.

Fourth, we explore domain adaptation more explicitly using Domain-Adversarial Neural Networks, or DANN. This allows us to investigate whether feature representations can be learned in a way that better generalizes across different domains, including combinations of database engine and hardware. We also examine semi-supervised domain adaptation settings, where only a small amount of labeled target-domain data is available. This is practically relevant because in real-world organizations, collecting fully labeled tuning data for every new server or database engine is expensive.

The broader significance of E2ETune++ lies in what it enables. If database tuning systems can generalize across engines and hardware, then AI-based optimization becomes far more usable in real operational environments. That matters especially for small and medium businesses, universities, non-profits, startups, and public sector institutions that lack access to specialist performance engineers. In these settings, infrastructure inefficiency is not just a technical inconvenience; it directly affects cost, service quality, and scalability.

E2ETune++ therefore represents more than a technical experiment. It is a software innovation that contributes to democratizing advanced infrastructure optimization. By reducing the expertise barrier required for effective database tuning, the project has the potential to make digital systems more efficient and accessible. Better tuning can reduce wasted compute resources, improve service responsiveness, and help organizations get more value from limited infrastructure budgets. In resource-constrained environments, that can have a meaningful social and economic impact.

The project is currently at the stage of a strong research-backed prototype with measurable experimental results and a clear path toward broader productization. It demonstrates that database tuning knowledge can be transferred across environments using AI, and that combining LLM-based tuning with transfer learning and cost modeling is a credible path toward more universal optimization systems.

E2ETune++ aligns strongly with the idea of “Tech for Good” because it applies advanced AI not to create novelty for its own sake, but to improve access to high-quality infrastructure optimization for organizations that would otherwise be excluded from it. With further development, validation, and deployment partnerships, it has the potential to become a practical, high-impact tool for more inclusive and efficient digital operations.

5.2 Explain How Your Innovation Solves a Problem

(Target: up to 500 words)

Draft Answer

E2ETune++ solves a practical and expensive problem that affects many organizations: modern databases are hard to tune well, and poor tuning leads to slower systems, wasted resources, and higher infrastructure cost.

In most real environments, database performance depends heavily on internal configuration parameters such as memory settings, I/O behavior, caching, concurrency, and execution-related knobs. The challenge is that these parameters interact in complex ways, and the best configuration depends on the workload, the database engine, and the hardware. As a result, good tuning typically requires experienced database administrators and many rounds of testing. Small organizations, universities, startups, and public institutions often do not have the resources or specialist expertise needed to do this effectively.

E2ETune++ addresses this by using AI to automate a large part of the tuning process. Instead of relying only on manual expert judgment or expensive repeated experiments, the system learns from workload features and performance data to recommend strong configurations. More importantly, it explores how this knowledge can be adapted across different environments, such as from PostgreSQL to MySQL and from one server type to another.

This matters because a solution that works only for one database engine or one machine is not enough in practice. Real organizations operate under changing constraints, diverse software stacks, and limited time. E2ETune++ moves toward a more usable model by combining cost prediction, workload clustering, transfer learning, and domain adaptation. This reduces the amount of new data that would otherwise need to be collected from scratch whenever the environment changes.

The “Tech for Good” value of the project lies in accessibility and efficiency. By lowering the expertise barrier for database optimization, E2ETune++ can help resource-constrained organizations improve their systems without needing highly specialized DBA support. Better tuning also means better use of compute resources, less performance waste, and more efficient digital infrastructure overall.

In short, E2ETune++ solves a high-value systems problem by turning advanced database optimization into something more transferable, scalable, and accessible. It helps bring the benefits of expert-level tuning closer to organizations that need efficiency the most but often have the fewest resources to achieve it.

5.3 Explain Why You Consider Your Solution Innovative

(Target: up to 500 words)

Draft Answer

E2ETune++ is innovative because it does not treat database tuning as an isolated optimization task. Instead, it combines recent advances in language models, cost modeling, transfer learning, and domain adaptation to address one of the biggest practical limitations in AI-driven infrastructure optimization: **generalization across environments**.

Many tuning approaches can perform reasonably well when trained and evaluated within a single controlled setting. The more difficult challenge is whether that knowledge remains useful when the database engine changes, when the hardware changes, or when only limited labeled data is available in the target environment. E2ETune++ focuses directly on that challenge.

The project is innovative in several ways. First, it extends an emerging LLM-based tuning paradigm rather than relying only on traditional search or manual rule systems. Second, it does not stop at direct model training; it introduces a supporting pipeline that includes workload embeddings, clustering, HEBO-based data generation, and learned cost models to make the full system more practical. Third, it explicitly investigates transfer learning from PostgreSQL to MySQL, which is a realistic and non-trivial cross-database challenge due to differences in behavior, performance distributions, and configuration spaces.

A particularly innovative part of the project is that it does not rely on one modeling philosophy. It tests and compares multiple adaptation strategies, including gradient-boosting-based regression transfer, residual correction methods, joint weighted training, and DANN-based domain-invariant feature learning. This gives the work more scientific and practical strength than a single-method prototype.

The project is also innovative in its intent. Rather than building a tool only for ideal lab conditions, E2ETune++ is designed around the reality that organizations have heterogeneous systems and limited resources. That makes the innovation socially meaningful as well as technically interesting.

Overall, E2ETune++ is innovative because it moves AI-based database tuning closer to real-world usability. It shows how advanced machine learning can help reduce the expertise barrier in database optimization, while also addressing the transfer problem that stands in the way of broader adoption.

5.4 Explain How Your Product Differs from Other Similar Products in the Market

(Target: up to 500 words)

Draft Answer

E2ETune++ differs from similar solutions in both technical approach and practical focus.

Traditional database tuning is usually done manually or through iterative search methods that require repeated live testing on the database system. These approaches can be effective, but they are costly in time, compute, and human expertise. They are also difficult to scale when workloads or environments change.

Earlier automated tuning systems such as classical Bayesian optimization frameworks mainly search for good configurations through repeated evaluation. They do not directly generalize well across environments, and they can still require substantial experimentation each time a new scenario is introduced.

More recent LLM-assisted approaches often use language models in a supporting role, such as reading documentation, narrowing the search space, or assisting with recommendation logic. By contrast, the E2ETune line of work treats the problem more directly as an end-to-end prediction task. E2ETune++ builds on this direction but adds something critical that many similar systems do not address sufficiently: **cross-environment adaptability**.

The main difference is that E2ETune++ is not only interested in generating database configurations; it is specifically designed to explore how tuning knowledge can transfer from one domain to another. In our project, that means from PostgreSQL to MySQL and from one hardware setting to another. This focus on domain adaptation makes the project more aligned with real operational diversity than systems trained only for a single fixed environment.

Another important difference is the layered pipeline. E2ETune++ combines representative workload clustering, cost modeling, transfer learning experiments, and LLM-based tuning. This gives it a broader systems perspective. It is not just "an LLM that outputs settings." It is a research-backed framework for reducing the cost of optimization and improving adaptability.

Finally, E2ETune++ differs in its potential accessibility value. Its long-term direction is not only to improve performance for already well-resourced teams, but also to reduce the optimization gap for smaller organizations that cannot afford specialist infrastructure support. That makes the project distinct not only as a technical system, but also as a socially relevant one.

5.5 Explain if Your Technology Is Part of an Emerging Area of Innovation

(Target: up to 500 words)

Draft Answer

Yes. E2ETune++ sits at the intersection of several emerging innovation areas.

The first is **LLM-powered systems optimization**. Language models are increasingly being used beyond text generation, including coding assistance, workflow automation, decision support, and configuration recommendation. Applying LLM-style fine-tuning to database knob tuning is part of this broader shift toward using generative AI for operational and systems-level intelligence.

The second is **AI for database management systems**, sometimes referred to as AI-driven database optimization. This is a rapidly evolving research and industry area because database performance directly affects cloud cost, application speed, scalability, and user experience. As digital services grow, the need for intelligent automation in infrastructure management becomes more important.

The third is **transfer learning and domain adaptation for systems problems**. Transfer learning is well established in computer vision and natural language processing, but its application to database tuning and cost prediction is still emerging. E2ETune++ contributes to this space by investigating how performance knowledge can move across database engines and hardware environments, rather than requiring complete retraining for every new case.

The fourth is **data-efficient optimization**. In many practical environments, it is expensive to collect large volumes of labeled system performance data. The project’s use of representative workload selection, learned cost models, and semi-supervised or cross-domain strategies reflects a broader move toward systems that can do more with less data.

The fifth is **sustainable and accessible AI infrastructure tooling**. There is increasing interest in tools that reduce wasted compute, improve efficiency, and make advanced infrastructure practices accessible to organizations with limited specialist capacity. E2ETune++ aligns with this direction because better tuning can improve resource usage and reduce the need for constant expert intervention.

For these reasons, E2ETune++ is clearly part of an emerging area of innovation. It brings together large language models, transfer learning, systems engineering, and practical optimization in a way that reflects where both research and industry are heading.

5.6 Explain How This Innovation Can Be Turned Into a Sustainable Product

(Target: up to 500 words)

Draft Answer

E2ETune++ has a credible path toward becoming a sustainable product because the underlying problem is persistent, widespread, and commercially relevant.

Almost every organization that operates a serious digital platform depends on database performance. However, many do not have the internal expertise needed to tune systems well. This creates a clear opportunity for a software product that can provide intelligent, explainable configuration recommendations without requiring specialist infrastructure teams.

A sustainable product direction for E2ETune++ would be a software platform or service that helps organizations optimize database settings based on workload characteristics, engine type, and hardware environment. In the near term, this could take the form of a decision-support tool for engineers and administrators. In the longer term, it could evolve into a cloud-based recommendation system, an internal DevOps support product, or a managed performance-optimization assistant.

Potential customer segments include:

- small and medium businesses
- startups running SaaS platforms
- universities and research labs
- managed service providers
- cloud and hosting providers
- public sector IT teams

A product version could be offered through multiple models:

- **Open-source core + paid support**
- **SaaS optimization dashboard**
- **Enterprise/on-prem deployment**
- **Consulting-assisted deployment for specialized environments**

The system also has strong expansion potential. Once the core methodology is validated, new database engines and hardware profiles can be added incrementally rather than requiring a complete redesign. This supports long-term scalability.

For sustainability, the product should also include explainability, auditability, and safe recommendation boundaries. Organizations are more likely to trust recommendations when they can understand why they were made and how they relate to workload behavior.

The commercial and operational sustainability of E2ETune++ would be further strengthened by partnerships with academic labs, cloud vendors, database tooling communities, or local enterprise software providers. If validated in real organizational settings, it could become a practical product that improves infrastructure efficiency while lowering the barrier to expert-level optimization.

5.7 Explain What Further Improvements Can Be Made to This Product

(Target: up to 500 words)

Draft Answer

E2ETune++ has several strong avenues for future improvement.

The first is **broader database support**. At present, the project focuses on PostgreSQL and MySQL-related adaptation challenges. A natural next step would be extending support to other systems such as MariaDB, SQL Server, or cloud-managed database services. This would improve product relevance and expand the number of organizations that could benefit.

The second is **better hardware-awareness and broader deployment conditions**. Real environments differ not only in RAM and CPU, but also in storage type, virtualization, containerization, and workload concurrency. Strengthening the model's robustness across these factors would make it more deployment-ready.

The third is **end-to-end productization**. The research system could evolve into a more user-friendly application with a dashboard, recommendation explanations, configuration comparison views, and a guided workflow for safe deployment. That would make the technology easier to adopt outside a research setting.

The fourth is **improved adaptation methods**. The project already explores multiple transfer and domain-adaptation strategies, but further work could improve stability, especially in adversarial learning settings. More robust semi-supervised learning, confidence-aware recommendation, and multi-stage adaptation could strengthen performance in low-data domains.

The fifth is **online or continual learning**. Instead of adapting only through offline training, future versions could learn from ongoing system feedback, gradually becoming better suited to a given environment over time.

The sixth is **multi-objective optimization**. Many real organizations care about more than latency alone. Future versions could jointly optimize throughput, responsiveness, resource efficiency, and cost.

The seventh is **real-world validation**. Pilot studies with university systems, startup applications, or institutional databases would provide stronger evidence of impact and guide product refinement.

Overall, E2ETune++ already demonstrates a compelling direction. Future development can make it broader, more stable, easier to use, and more directly deployable in real operational environments.

5.8 Explain Efforts That Have Been Made to Raise Awareness of Your Product

(Target: up to 500 words)

Draft Answer

E2ETune++ is currently positioned as a research-driven university innovation, and awareness-building has primarily focused on technical validation, documentation, and academic presentation.

The project has been developed through structured supervision, repeated technical reviews, experimental comparison, and refinement of both methodology and evaluation. Throughout the project, the team has actively discussed system design, cost modeling strategies, workload selection, transfer learning methods, and domain adaptation findings in formal review settings. This has helped shape the work into a more mature and defensible innovation.

The team has also built awareness through technical documentation and reproducibility-oriented assets such as code repositories, experimental outputs, architecture discussions, and benchmark-based evaluation materials. These materials form an important base for broader awareness because they show that the project is not just conceptual, but implementation-backed and evidence-driven.

Where available, the following awareness activities can be highlighted and linked as supporting evidence:

- project presentations at **[Insert university name]**
- departmental reviews or research presentations **[Insert evidence if available]**
- GitHub repository or technical documentation **[Insert GitHub link]**
- poster/demo sessions **[Insert evidence if available]**
- short explanatory video or prototype showcase **[Insert link]**
- faculty or external reviewer feedback **[Insert evidence if available]**

Going forward, the team intends to raise awareness more broadly by publishing a public-facing project summary, sharing results with database and AI communities, participating in university innovation showcases, and seeking pilot opportunities with organizations that can benefit from more accessible optimization support.

As a “Tech for Good” innovation, awareness-building is not only about publicity; it is about communicating why this problem matters to organizations that are often underserved in advanced infrastructure tooling. E2ETune++ aims to contribute to that conversation by showing how AI can improve access to efficient digital infrastructure, not just for elite engineering teams, but for a wider range of institutions.

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6. Additional Draft for “Best Innovative Product - University”

> This category is likely the stronger fit. These answers should emphasize **student-led innovation, technical originality, real-world relevance, feasibility, and future potential**.

6.1 Provide a Detailed Write-Up on Your Innovative Product/Project

(Target: up to 1000 words)

Draft Answer

E2ETune++ is a university final year project that explores how artificial intelligence can make database performance optimization more intelligent, transferable, and accessible. The project focuses on a real problem in modern computing: databases are central to digital systems, but tuning them well requires specialist expertise, repeated testing, and substantial computing effort.

Databases such as PostgreSQL and MySQL expose many configuration parameters that strongly affect performance. These settings influence memory usage, query execution, I/O efficiency, concurrency, caching, and related behavior. In practice, selecting good values is difficult because the best configuration depends not only on the database engine, but also on workload characteristics and the underlying hardware. As a result, many organizations either rely on scarce experts or continue to operate with default or suboptimal settings.

Our project was inspired by E2ETune, a recent research direction that uses a language-model-based approach to map workload information directly to promising database configurations. We extended this idea in a more practical direction by asking a harder question: can an AI tuning system still work when we move across different database engines and different server environments?

To address this, we developed E2ETune++, a research-backed prototype that combines multiple technical components into one coherent pipeline.

First, we use workload embeddings and clustering to make data collection more efficient. Instead of executing every available workload scenario, we normalize SQL workloads, generate embeddings, and use HDBSCAN clustering to identify representative subsets. This helps us reduce redundancy while preserving workload diversity across standard benchmarks.

Second, we use HEBO-based Bayesian optimization to generate valuable configuration-performance data from benchmark executions. These results are then used to train cost models that predict database latency or performance from workload, configuration, hardware, and database-related features.

Third, we investigate transfer learning from PostgreSQL to MySQL. This is important because PostgreSQL data is more abundant in our project, while MySQL data is more limited. Rather than discarding that imbalance as a problem, we treat it as a research opportunity. We test whether models trained with richer PostgreSQL data can help improve prediction on MySQL using techniques such as residual adapters and joint weighted training.

Fourth, we explore domain-adversarial learning through DANN to investigate whether more domain-invariant feature representations can be learned across combinations of database engine and hardware. We also study semi-supervised adaptation settings where only a small amount of labeled target-domain data is available.

The result is a project that is not only technically ambitious but also practically motivated. E2ETune++ is a strong example of interdisciplinary innovation, combining machine learning, language models, transfer learning, optimization, and database systems engineering. As an undergraduate project, it demonstrates originality in problem framing, depth in experimentation, and strong awareness of real-world constraints.

What makes the project especially meaningful is that it aims to reduce the expertise barrier in infrastructure optimization. If advanced tuning can become more portable and more automated, then the benefits of good performance engineering can reach smaller and less-resourced organizations as well.

E2ETune++ is therefore both a serious technical project and a promising foundation for future research and product development. It demonstrates that university-level innovation can contribute meaningfully to current challenges at the frontier of AI and systems engineering.

6.2 Explain the Problem Your Product/Project Solves and Its Impact

(Target: up to 500 words)

Draft Answer

E2ETune++ addresses the problem of difficult, expensive, and environment-specific database tuning.

Database systems are essential to web platforms, enterprise software, public services, educational tools, and analytics applications. However, good performance depends heavily on internal configuration settings. Poor tuning can lead to slower applications, inefficient resource usage, and higher infrastructure cost. The challenge is that these configurations are highly dependent on workload patterns, database engine behavior, and hardware characteristics.

In practice, effective tuning often requires specialist knowledge that many organizations do not have. Smaller organizations, universities, public institutions, and startups are particularly affected because they cannot always afford dedicated database performance experts. This creates a gap between what is technically possible and what is practically available.

E2ETune++ addresses this problem by using AI to reduce the cost and expertise needed for database optimization. It explores how a tuning system can learn from workloads and recommend configurations, while also adapting knowledge across different database engines and hardware environments. This is important because real organizations rarely operate in a perfectly controlled single-environment setting.

The impact of this work is twofold. Technically, it contributes to more efficient and transferable optimization methods. Practically, it moves toward making advanced infrastructure tuning more accessible to organizations with limited specialist capacity. Better database tuning can improve responsiveness, reduce waste, and help organizations use their existing hardware more effectively.

As a university project, E2ETune++ shows that student-led research can address a meaningful systems problem with both technical and broader societal value.

6.3 Describe Why Your Product/Project Is Innovative

(Target: up to 500 words)

Draft Answer

E2ETune++ is innovative because it extends an emerging LLM-based database tuning paradigm into a more realistic cross-domain setting.

The project does not simply apply one model to one dataset. Instead, it combines workload clustering, cost modeling, Bayesian data generation, transfer learning, and domain adaptation to address a difficult problem: how to make tuning knowledge portable across database engines and hardware settings.

This is innovative in both method and perspective. Methodologically, the project explores several advanced approaches, including LightGBM-based transfer learning, residual correction, weighted multi-domain training, and DANN-based adversarial adaptation. Conceptually, it treats database tuning not as a one-time optimization problem, but as a generalization problem.

The project is also innovative because it aims to reduce the amount of new data needed when moving to a new environment. That makes the research more aligned with practical deployment constraints.

Finally, the project is innovative as an undergraduate effort because it brings together ideas from multiple advanced areas of computing—databases, machine learning, language models, optimization, and systems engineering—and applies them in a coherent way to solve a real infrastructure problem.

6.4 Highlight What Sets Your Product/Project Apart from Existing Alternatives

(Target: up to 500 words)

Draft Answer

What sets E2ETune++ apart is its focus on **adaptability**.

Many database tuning approaches are either manual, highly search-based, or tightly tied to a single environment. E2ETune++ differs by exploring whether tuning intelligence can transfer across environments. This makes it more relevant to real-world scenarios where organizations use different database engines, different hardware, and limited data.

The project also stands out because it integrates multiple components rather than relying on a single model. It includes workload selection through embeddings and clustering, cost model training, transfer learning experiments, and LLM-based end-to-end tuning ideas. This gives it a stronger systems perspective than a narrow modeling-only approach.

Another key differentiator is that the project explicitly examines data imbalance and low-data target settings. In practical deployment, organizations often do not have large, clean, well-labeled tuning datasets for every environment. E2ETune++ addresses this challenge directly.

As a student project, it also stands out for its ambition and depth. It is not just a prototype interface or a basic machine learning model. It is a structured attempt to advance the state of intelligent database optimization in a way that is both technically credible and potentially useful.

6.5 Detail the Technology Used in Development

(Target: up to 500 words)

Draft Answer

E2ETune++ uses a combination of modern AI, machine learning, and systems engineering technologies.

At the modeling level, the project follows an **LLM-based fine-tuning approach inspired by Mistral-7B-style models**, based on the E2ETune direction for end-to-end database knob recommendation. For workload representation, SQL workloads are normalized and embedded using **gemini-embedding-001**, then clustered using **HDBSCAN** to identify representative workload groups.

For data generation, the project uses **HEBO**, a Bayesian optimization framework, to explore configuration space and collect useful configuration-performance data from benchmark executions. This data is used to train **cost models** that estimate latency or related performance indicators.

The cost modeling experiments use **LightGBM**, **RandomForest**, and **HistGradientBoosting**. For transfer learning from PostgreSQL to MySQL, the project evaluates a **MySQL-only baseline**, a **Residual Adapter** approach, and **Joint Weighted Training**.

The project also explores **Domain-Adversarial Neural Networks (DANN)** implemented in a neural architecture with a feature extractor, label predictor, and domain classifier connected through a gradient reversal layer. This supports experiments in domain-invariant feature learning and semi-supervised adaptation.

The benchmark suite includes well-known OLAP and OLTP workloads such as JOB, SSB, TPC-H, TPC-DS, SMALLBANK, TPCC, TWITTER, WIKIPEDIA, YCSB, and SSB_FLAT. The hardware settings span multiple server profiles, including 64 GB and 32 GB RAM environments with different CPU and storage characteristics.

Overall, the development stack combines language-model-driven reasoning, classical machine learning, neural domain adaptation, workload representation, and benchmark-based systems evaluation.

6.6 State Whether Your Technology Falls Under an Emerging Field and Its Significance

(Target: up to 500 words)

Draft Answer

Yes. E2ETune++ falls under several emerging fields, most notably AI-driven systems optimization, LLM applications beyond text generation, and transfer learning for infrastructure engineering.

One of the most significant trends in computing today is the use of large language models for structured problem-solving tasks beyond ordinary language use. E2ETune++ fits this direction by applying an LLM-style fine-tuning approach to database configuration recommendation.

The project also belongs to the growing field of **AI for database systems**, where machine learning is used to improve tuning, optimization, resource allocation, and performance management. As digital systems grow more complex, this area is becoming increasingly important for both research and industry.

A further emerging dimension is the application of **transfer learning and domain adaptation** to systems problems. While transfer learning is common in vision and language tasks, its use in database tuning remains relatively new. E2ETune++ is significant because it investigates whether tuning knowledge can be reused across engines and hardware domains, which is a highly practical problem.

The significance of this field lies in its potential to reduce manual infrastructure effort, increase system efficiency, and make advanced optimization more widely available. That is why E2ETune++ is not only technically current, but also aligned with an important future direction in computing.

6.7 Explain How This Can Be Commercialized

(Target: up to 500 words)

Draft Answer

E2ETune++ has commercialization potential as an intelligent database optimization assistant for organizations that want better performance without needing deep in-house tuning expertise.

A practical commercialization path could begin with a decision-support product that accepts workload and environment information and recommends promising database configurations. This could be delivered as:

- a self-hosted enterprise tool,
- a managed SaaS platform,
- a plugin for DevOps/database administration workflows, or
- a consulting-assisted platform for high-value deployments.

Potential customers include SMEs, SaaS startups, managed service providers, educational institutions, and public sector IT teams. These groups often have clear performance needs but limited access to specialist DBA resources.

The value proposition is strong: better performance, more efficient resource usage, reduced trial-and-error effort, and faster tuning guidance. Over time, the system could expand to support more database engines, workload types, and deployment environments.

Commercialization would be strengthened by features such as explainable recommendations, rollback-safe tuning suggestions, deployment policies, and support for popular database platforms. With additional validation and interface refinement, E2ETune++ could develop into a credible infrastructure optimization product.

6.8 Outline Potential Improvements and Future Enhancements

(Target: up to 500 words)

Draft Answer

Future improvements for E2ETune++ include technical, usability, and deployment-oriented enhancements.

On the technical side, the project can be expanded to support more database engines and broader hardware variation. More robust adaptation strategies, improved cost models, and more stable semi-supervised learning pipelines would also strengthen generalization.

On the product side, the project would benefit from a more polished interface, better explainability, and guided deployment workflows that help users understand and safely apply recommendations. Monitoring and feedback loops could also allow the system to refine recommendations over time.

A particularly important enhancement would be real-world validation through pilot deployments. This would help the team measure impact more directly and improve the product based on actual user needs.

Longer term, E2ETune++ could evolve toward continual learning, multi-objective tuning, and broader integration into cloud or enterprise infrastructure stacks. These improvements would move it from a strong research prototype toward a deployable optimization platform.

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7. Evidence Checklist to Strengthen the Application

> **Gather these before final submission.**

Core Eligibility / University Proof

- [ ] **Supervisor or Department Head confirmation letter**
- [ ] **Proof of undergraduate status**
- [ ] **Team member list and roles**
- [ ] **Proof of originality / ownership statement**

Technical Evidence

- [ ] **Overall architecture diagram**
- [ ] **DANN architecture diagram**
- [ ] **Data generation / training pipeline diagram**
- [ ] **Benchmark list with short explanations**
- [ ] **Hardware setup table**
- [ ] **Cost model results tables**
- [ ] **Transfer learning results tables**
- [ ] **DANN ablation visuals**
- [ ] **Screenshots of database benchmark executions**
- [ ] **Screenshots of any UI/demo/prototype**
- [ ] **GitHub repository link**
- [ ] **Short technical appendix or README**
- [ ] **Explainability diagrams / flowcharts**

Submission Assets

- [ ] **5-minute video link**
- [ ] **4-page brochure PDF**
- [ ] **Slide deck PDF**
- [ ] **Product screenshots**
- [ ] **Team photo and project branding assets**

Impact and Credibility Evidence

- [ ] **Why this is Tech for Good justification**
- [ ] **Any pilot testing or informal user feedback**
- [ ] **Any faculty, reviewer, or industry feedback**
- [ ] **Any presentation/demo event evidence**
- [ ] **Any publication or submission evidence if available**
- [ ] **Any reproducibility or open-source evidence**

Final Sanity Check

- [ ] All metrics verified
- [ ] No unsupported impact claims
- [ ] Category wording tailored correctly
- [ ] All placeholders replaced
- [ ] Links tested
- [ ] English polished and consistent

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8. Judge-Friendly Results Summary

8.1 What These Results Mean in Plain Language

These results show three important things:

1. **The team built strong cost prediction models for PostgreSQL**
2. **Knowledge from PostgreSQL can help improve MySQL prediction significantly**

3. **Advanced domain adaptation is promising, but some approaches are still less stable than simpler supervised transfer methods**

That is actually a strong story. It shows both **technical ambition** and **honest, evidence-based learning**.

8.2 PostgreSQL-Only Regression Results

Model	RMSE	MAPE	Spearman ρ
LightGBM	**0.9710 ± 0.2046**	**5.54 ± 1.21%**	0.9417 ± 0.0154
RandomForest	1.0074 ± 0.2921	5.77 ± 1.20%	**0.9532 ± 0.0171**
HistGradientBoosting	1.0429 ± 0.2782	5.85 ± 1.03%	0.9244 ± 0.0125

Judge-Friendly Interpretation

- All three models perform strongly
- Errors are relatively low
- Spearman values above 0.92 indicate the models are **very good at ranking configurations**
- This means the project has a solid predictive foundation for tuning support

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8.3 MySQL Adaptation Results

Method	RMSE	MAPE	Spearman ρ
MySQL-Only Regressor	3.3853 ± 0.6655	12.26 ± 5.85%	0.4693 ± 0.0437
Residual Adapter	3.3402 ± 0.6729	5.93 ± 0.73%	0.5681 ± 0.0431
Joint Weighted Training	**3.1431 ± 0.5958**	**4.81 ± 0.23%**	**0.6118 ± 0.0359**

Judge-Friendly Interpretation

- Training only on the small MySQL dataset gives weaker performance
- Transfer learning from PostgreSQL helps significantly
- **Joint Weighted Training is the best performer**
- MAPE improves from **12.26% to 4.81%**
- This is one of the strongest findings in the whole project

Simplified Takeaway

> **The team showed that knowledge learned from a richer PostgreSQL setting can materially improve predictions for a smaller MySQL setting.**

That is exactly the kind of result judges can understand and value.

8.4 DANN Ablation Summary

DANN Study Context

Item	Value
Samples	**10,190**
Domains	**4**
Raw Features	**79**
Encoded Dimensions	**447**
Model Parameters	**548,165**

Best Mean Results Reported

Normalization Strategy	Best Setup	RMSE	NRMSE	MAPE	Spearman ρ
Global Normalization	**SSDA 10%**	**45.69**	**7.58%**	246.18%	0.631
Per-Engine Normalization	**Supervised**	46.37	8.54%	**99.90%**	**0.843**

Judge-Friendly Interpretation

- DANN explored a harder and more experimental adaptation setting
- Results show that **normalization strategy matters a lot**
- The best adversarial/semi-supervised setup improved some metrics
- But in practical terms, **simpler supervised learning with per-engine normalization performed more reliably**
- This is still valuable because it reveals what works and what does not

Simplified Takeaway

> **The project did not just chase a complex method. It rigorously tested whether domain-adversarial learning helps, and found that simpler supervised transfer may currently be more stable in this setting.**

That is a strong research message, not a weakness.

8.5 Best Way to Present the Results to Judges

Use this wording:

- **"Our results show that cross-database transfer learning works."**
- **"We achieved strong predictive accuracy on PostgreSQL and significantly improved MySQL prediction through transfer learning."**
- **"Our ablation studies helped us identify which adaptation strategies are most stable in practice."**
- **"This means E2ETune++ is moving toward practical generalization across real-world environments."**

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9. Five-Minute Video / Presentation Master Content

> Recommended length: **14 slides**
> Recommended pace: **20–25 seconds per slide**
> Tone: confident, clear, practical, judge-friendly

Slide 1 — Title

Objective

Introduce the project clearly and professionally.

On-Slide Content

- **E2ETune++**
- **Domain-Adapted End-to-End Database Knob Tuning Using LLMs and Transfer Learning**
- **SLASSCOM National Ingenuity Awards 2026**
- **[Insert University Name]**

- **[Insert Team Names]**

Speaker Notes

“Hello, we are [team name] from [university]. Our project, E2ETune++, is an AI-driven database tuning system that uses large language models, transfer learning, and domain adaptation to make advanced database optimization more accessible across different database engines and hardware environments.”

Recommended Visual

Professional title slide with:

- database icon
- AI/brain motif
- subtle performance/dashboard theme

Slide 2 — The Problem

Objective

Make judges care immediately.

On-Slide Content

- Databases power critical digital systems
- Performance depends on many complex internal settings
- Good tuning requires specialist expertise
- Poor tuning leads to:
  - slower systems
  - wasted compute resources
  - higher infrastructure cost
- Most smaller organizations cannot afford expert DBA support

Speaker Notes

“Database performance is critical, but tuning databases properly is difficult. The right configuration depends on workload, engine behavior, and hardware. In practice, expert tuning is expensive and time-consuming, and many organizations continue running on inefficient settings.”

Recommended Visual

Problem infographic:

- slow app
- server waste
- costly expertise
- complex knobs

Slide 3 — Why This Matters

Objective

Link the technical problem to broader value.

On-Slide Content

- Impacts startups, universities, SMEs, and public institutions
- Affects speed, scalability, cloud cost, and service quality
- Under-resourced organizations are hit hardest
- Need: **accessible, intelligent optimization**

Speaker Notes

“This is not only a technical inconvenience. It affects how efficiently organizations run digital services. For smaller or under-resourced teams, the lack of tuning expertise can directly limit service quality and increase cost.”

Recommended Visual

Audience impact diagram:

- startup
- university
- SME
- government IT unit

Slide 4 — Our Solution

Objective

Present E2ETune++ simply.

On-Slide Content

- AI-driven database tuning system
- Extends the E2ETune research direction
- Combines:
  - LLM-based tuning
  - workload clustering
  - cost modeling
  - transfer learning
  - domain adaptation
- Goal: adapt across PostgreSQL, MySQL, and different hardware setups

Speaker Notes

“E2ETune++ extends an emerging LLM-based tuning approach and adds what real-world systems need most: the ability to adapt across databases and hardware settings instead of working only in one fixed environment.”

Recommended Visual

High-level system block diagram

Slide 5 — How the System Works

Objective

Explain the pipeline in one slide.

On-Slide Content

Pipeline

1. Collect representative workloads
2. Generate performance data with HEBO
3. Train cost models
4. Transfer knowledge across domains
5. Fine-tune LLM-based recommendation pipeline

Speaker Notes

“Our system first selects representative workloads, then collects optimization data using HEBO, then trains cost models to predict performance, and finally explores how tuning knowledge can transfer across different environments.”

Recommended Visual

5-step horizontal flowchart

Slide 6 — Workload Clustering Innovation

Objective

Show smart data efficiency.

On-Slide Content

- SQL normalization
- Embeddings with **gemini-embedding-001**
- Clustering using **HDBSCAN**
- Goal: choose representative workloads instead of running everything
- Benchmarks include JOB, SSB, TPC-H, TPC-DS, SMALLBANK, TPCC, TWITTER, WIKIPEDIA, YCSB, SSB_FLAT

Speaker Notes

“One key innovation is efficient workload selection. Instead of brute-forcing all workloads, we cluster them and select representative ones. This reduces unnecessary experimentation while preserving workload diversity.”

Recommended Visual

Cluster map or stylized grouped workload bubbles

Slide 7 — Cost Model Foundation

Objective

Show the base system is solid.

On-Slide Content

PostgreSQL dataset

- 217 workloads
- 20,758 samples

Models tested

- LightGBM
- RandomForest
- HistGradientBoosting

Best results

- LightGBM MAPE: **5.54%**
- RandomForest Spearman: **0.9532**

Speaker Notes

"We first built strong PostgreSQL cost models. The results were consistently strong, showing that the system can accurately estimate performance and rank promising configurations."

Recommended Visual

Bar chart of PostgreSQL model performance

Slide 8 — Cross-Database Transfer Learning

Objective

Highlight the strongest result.

On-Slide Content

Challenge

- PostgreSQL data-rich
- MySQL data-scarce

MySQL dataset

- 46 workloads
- 1,314 samples

Adaptation methods

- MySQL-only
- Residual Adapter
- Joint Weighted Training

Best result

- Joint Weighted Training MAPE: **4.81%**
- vs MySQL-only MAPE: **12.26%**

Speaker Notes

"Our strongest finding is that transfer learning helps significantly. Instead of relying only on limited MySQL data, we used PostgreSQL knowledge to improve MySQL prediction quality. Joint weighted training produced the best result."

Recommended Visual

Before vs after comparison chart

Slide 9 — DANN and Domain Adaptation

Objective

Show research depth and advanced experimentation.

On-Slide Content

- Explored **Domain-Adversarial Neural Networks**
- 10,190 samples
- 4 domains
- 79 features / 447 encoded dimensions
- Tested:
  - source-only
  - semi-supervised adaptation
  - supervised baseline
- Key finding: simpler supervised transfer was more stable in this setting

Speaker Notes

"We also explored a more advanced domain-adversarial approach using DANN. This helped us study domain invariance across engine and hardware settings. The findings were valuable: some advanced setups were promising, but simpler supervised transfer remained more stable."

Recommended Visual

DANN architecture diagram with GRL

Slide 10 — Hardware Awareness

Objective

Show realism.

On-Slide Content

Server 1

- Xeon E3-1275v5
- 64 GB RAM
- NVMe SSD

Server 2

- Core i7-7700
- 32 GB RAM
- SATA SSD

Goal

- move beyond single-hardware assumptions

Speaker Notes

"A major practical issue in database tuning is that performance recommendations often do not generalize across hardware. Our work explicitly considers different server profiles rather than assuming one ideal environment."

Recommended Visual

Side-by-side server cards

Slide 11 — Why This Is Tech for Good

Objective

Make the social-good case explicit.

On-Slide Content

- Reduces dependence on scarce specialist DBAs
- Helps resource-constrained organizations
- Improves efficiency of digital infrastructure
- Reduces waste from poor configurations
- Supports more inclusive access to advanced optimization

Speaker Notes

“This is Tech for Good because it aims to make advanced performance optimization more accessible. Better tuning should not be limited to organizations that can afford specialist infrastructure teams. E2ETune++ helps close that gap.”

Recommended Visual

Impact infographic with 4 icons:

- accessibility
- efficiency
- affordability
- inclusion

Slide 12 — Why It Stands Out

Objective

Summarize uniqueness.

On-Slide Content

- Extends state-of-the-art LLM tuning research
- Combines LLMs, clustering, cost modeling, and transfer learning
- Addresses cross-database and cross-hardware generalization
- Backed by measurable experimental results
- Strong foundation for future productization

Speaker Notes

“What makes E2ETune++ stand out is that it combines advanced AI with a very practical systems challenge. It is not only about generating configurations; it is about making tuning knowledge portable and more usable in the real world.”

Recommended Visual

Feature comparison matrix vs conventional methods

Slide 13 — Future Direction

Objective

Show growth potential.

On-Slide Content

- More database engine support
- Better hardware generalization
- Real-world pilot deployments
- User-facing dashboard and explainability
- Safer automated recommendation workflow

Speaker Notes

“This project already demonstrates strong technical feasibility, and the next step is to broaden support, improve stability, and validate it in real environments through pilot deployment and user-facing tooling.”

Recommended Visual

Roadmap timeline

Slide 14 — Closing

Objective

End memorably.

On-Slide Content

E2ETune++

- Makes advanced tuning more accessible
- Shows real cross-domain learning potential
- Combines technical depth with practical value
- A university project with strong future impact

Speaker Notes

“In summary, E2ETune++ is a strong example of how AI can be used to solve a real infrastructure problem in a more inclusive way. It combines technical innovation with practical relevance, and we believe it represents the kind of systems-focused, socially meaningful innovation that deserves recognition.”

Recommended Visual

Strong branded closing slide with tagline:
“Making expert-level database optimization more accessible.”

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10. Exact 5-Minute Voiceover Script

Hello, we are **[team name]** from **[university name]**, and we are presenting **E2ETune++: Domain-Adapted End-to-End Database Knob Tuning Using LLMs and Transfer Learning**.

Modern digital systems depend on databases. But getting good database performance is difficult because databases contain many internal configuration parameters, often called knobs, that affect memory usage, I/O behavior, concurrency, caching, and query execution. Setting these correctly requires significant expertise, repeated experimentation, and time. In many organizations, especially startups, universities, SMEs, and public institutions, that expertise is either limited or unavailable.

This is the problem our project addresses.

E2ETune++ is an AI-driven database tuning system designed to make advanced optimization more accessible and more adaptable across real environments. It builds on the E2ETune research direction, where a language-model-based system learns to recommend promising database configurations from workload information. Our key contribution is that we do not assume one fixed database engine or one fixed server. Instead, we explore how tuning knowledge can transfer across **PostgreSQL and MySQL**, and across **different hardware settings**.

Our system combines several components.

First, we use **workload embeddings and HDBSCAN clustering** to identify representative workloads, so we do not need to execute every possible scenario. This helps us collect data more efficiently across major benchmark suites such as JOB, SSB, TPC-H, TPC-DS, SMALLBANK, TPCC, TWITTER, WIKIPEDIA, YCSB, and SSB_FLAT.

Second, we use **HEBO-based Bayesian optimization** to generate high-value performance data. This data is used to train **cost models** that predict workload latency and performance from workload features, knob settings, hardware details, and database type.

Third, we investigate **transfer learning** from a large PostgreSQL dataset to a much smaller MySQL dataset. This is one of our strongest findings. Our PostgreSQL experiments included 217 workloads and 20,758 samples. For MySQL, we had only 46 workloads and 1,314 samples. We tested multiple regression-based approaches, including a MySQL-only baseline, a Residual Adapter, and Joint Weighted Training.

The results clearly show the value of transfer learning. The MySQL-only model had a MAPE of **12.26%**, while the Residual Adapter reduced that to **5.93%**, and Joint Weighted Training reduced it further to **4.81%**. This shows that knowledge learned from a richer PostgreSQL environment can significantly improve prediction in a smaller MySQL setting.

We also explored **Domain-Adversarial Neural Networks**, or DANN, to study domain adaptation more deeply across combinations of database engine and hardware. This involved 10,190 samples, 4 domains, and a neural architecture with over 548 thousand parameters. These experiments showed that domain adaptation is promising, but also that simpler supervised transfer methods can currently be more stable in practice. That is an important and honest research finding.

From a practical perspective, our project matters because most organizations do not operate in a single ideal environment. Hardware changes. Database engines differ. Data availability is uneven. E2ETune++ addresses that real-world complexity.

This is also why we believe E2ETune++ fits strongly under **Tech for Good**. Better database tuning should not be limited to organizations with specialist DBA teams. By reducing the expertise barrier, our project can help resource-constrained organizations improve performance, reduce inefficiency, and make better use of limited infrastructure. In that sense, this is not only a technical innovation, but also an accessibility and efficiency innovation.

As a university final year project, E2ETune++ demonstrates strong originality, serious experimentation, and clear future potential. It brings together large language models, workload clustering, cost prediction, transfer learning, and systems engineering into a coherent solution for a real and costly problem.

In summary, E2ETune++ is a research-backed AI system that aims to make advanced database optimization more transferable, more efficient, and more accessible. We believe it represents the kind of technically strong and socially meaningful innovation that deserves recognition at the SLASSCOM National Ingenuity Awards.

Thank you.

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11. Product Brochure Content (4-page max)

Page 1 — Cover Page

Headline

E2ETune++

Subheadline

Domain-Adapted End-to-End Database Knob Tuning Using LLMs and Transfer Learning

Tagline

Making expert-level database optimization more accessible

Cover Copy

E2ETune++ is an AI-driven database tuning system that combines large language model-based recommendation, workload clustering, cost modeling, transfer learning, and domain adaptation to improve database optimization across PostgreSQL, MySQL, and different hardware environments.

Footer

- **[Insert University Name]**
- **[Insert Team Names]**
- **[Insert Contact Email]**
- **[Insert QR Code to GitHub or Demo]**

Page 2 — Problem and Solution

The Problem

Modern digital services rely on databases, but database performance depends heavily on complex internal configuration settings. Good tuning requires deep expertise, repeated testing, and significant time. Many organizations cannot afford specialist database administrators and continue running on inefficient settings.

Why This Matters

Poor tuning can lead to:

- slower systems
- inefficient resource usage
- higher infrastructure costs
- limited scalability
- unnecessary operational complexity

Our Solution

E2ETune++ is an AI-driven tuning framework that:

- analyzes workload characteristics
- predicts performance using trained cost models

- uses transfer learning across database environments
- explores domain adaptation across engines and hardware
- supports the broader goal of accessible, data-driven configuration recommendation

Key Value Proposition

From expensive, expertise-heavy tuning to intelligent, transferable optimization support

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Page 3 — Technical Innovation and Results

Technical Highlights

- **LLM-based tuning pipeline** inspired by E2ETune
- **Workload embeddings + HDBSCAN clustering**
- **HEBO-based data generation**
- **Cost models for latency/performance prediction**
- **Transfer learning from PostgreSQL to MySQL**
- **DANN-based domain adaptation experiments**

Key Results

PostgreSQL Cost Modeling

Model	RMSE	MAPE	Spearman ρ
LightGBM	**0.9710 ± 0.2046**	**5.54 ± 1.21%**	0.9417 ± 0.0154
RandomForest	1.0074 ± 0.2921	5.77 ± 1.20%	**0.9532 ± 0.0171**
HistGradientBoosting	1.0429 ± 0.2782	5.85 ± 1.03%	0.9244 ± 0.0125

MySQL Adaptation

Method	RMSE	MAPE	Spearman ρ
MySQL-only	3.3853 ± 0.6655	12.26 ± 5.85%	0.4693 ± 0.0437
Residual Adapter	3.3402 ± 0.6729	5.93 ± 0.73%	0.5681 ± 0.0431
Joint Weighted Training	**3.1431 ± 0.5958**	**4.81 ± 0.23%**	**0.6118 ± 0.0359**

What This Means

E2ETune++ shows that tuning knowledge can be transferred from data-rich database environments to lower-data ones, improving accessibility and reducing the need to start from scratch.

Page 4 — Impact, Roadmap, and Contact

Why It Matters

E2ETune++ aims to make advanced database optimization more accessible to:

- SMEs
- startups
- universities
- public institutions
- under-resourced technology teams

Tech for Good Value

- reduces dependency on scarce specialist expertise
- supports more efficient infrastructure usage
- reduces waste from poor configurations
- helps smaller organizations benefit from advanced AI-based optimization

Roadmap

- support more database engines
- improve hardware generalization
- build a user-facing recommendation dashboard
- add explainability and safer deployment workflows
- validate through pilot deployments

Contact

- **Team:** [Insert team names]
- **University:** [Insert university name]
- **Email:** [Insert email]
- **GitHub:** [Insert GitHub link]
- **Demo/Video:** [Insert link]

---

12. Diagram Guide

12.1 Overall E2ETune++ Architecture

Purpose

Show the full system at a glance.

What to Include

Left Side: Inputs

Three input boxes:

- **Workload Features**
- **Database Engine**
- **Hardware Profile**

Under workload features, optionally list:

- normalized SQL
- plan embeddings
- workload metrics

Center: E2ETune++ Core

Main large box titled **E2ETune++**
Inside it, show sub-boxes:

- **Workload Clustering**
- **HEBO Data Generation**
- **Cost Models**
- **Transfer Learning Module**
- **LLM-Based Recommendation Module**

Right Side: Outputs

Two output boxes:

- **Predicted Performance**
- **Recommended Configuration**

Arrows

- Inputs flow into core
- Clustering feeds data generation
- Data generation feeds cost models
- Cost models and transfer learning feed LLM-based recommendation
- Recommendation produces output

12.2 DANN Architecture with GRL

Purpose

Explain domain-adversarial learning clearly.

What to Include

Left

Input box: **Encoded Workload + Configuration + Environment Features**

Middle

Box: **Feature Extractor**

From Feature Extractor, split into two branches:

Upper Branch

- Box: **Label Predictor**
- Output: **Predicted Latency / Cost**

Lower Branch

- Box: **Gradient Reversal Layer (GRL)**
- Then box: **Domain Classifier**
- Output: **Predicted Domain**

Labels

- On upper branch: **Task Loss**
- On lower branch: **Domain Loss**

Note

Add a caption:
GRL reverses gradient direction to encourage domain-invariant feature learning

---

12.3 Data Generation and Fine-Tuning Pipeline

Purpose

Show how data becomes intelligence.

What to Include

1. **Benchmark Workloads**
2. **SQL Normalization**
3. **Embedding Generation**
4. **HDBSCAN Clustering**
5. **Representative Workload Selection**
6. **HEBO Optimization Runs**
7. **Collected Performance Data**
8. **Cost Model Training**
9. **Transfer / Adaptation Layer**
10. **LLM Fine-Tuning**
11. **End-to-End Recommendation**

Use arrows between each block.

At the bottom, add a side-note box:
Goal: reduce live experimentation while improving cross-domain tuning capability

---

12.4 Cross-Database Adaptation Approaches

Purpose

Compare the methods visually.

What to Include

Left Side

Two data boxes:

- **PostgreSQL (20,758 samples)**
- **MySQL (1,314 samples)**

Middle: Three adaptation paths

Path 1

- MySQL-only baseline
- arrow to model
- arrow to output

Path 2

- PostgreSQL base model
- residual correction module
- MySQL prediction

Path 3

- merged training data
- weighted learning block
- unified predictor

Right Side

Results table snippet:

- MySQL-only: 12.26% MAPE
- Residual Adapter: 5.93% MAPE
- Joint Weighted: 4.81% MAPE

Add highlight:
Best: Joint Weighted Training

12.5 Impact Infographic

Purpose

Support Tech for Good positioning.

What to Include

Center title:
Why E2ETune++ Matters

Four surrounding impact boxes:

1. **Accessibility**
   - reduces expertise barrier
2. **Efficiency**
   - better use of infrastructure
3. **Affordability**
   - helps smaller organizations
4. **Scalability**
   - supports adaptation across environments

Optional bottom banner:
AI for more inclusive and efficient digital infrastructure

---

13. Submission Pitfalls to Avoid

Do Not Overclaim

- Do **not** claim large-scale production deployment unless true
- Do **not** say the project is already widely used unless you have evidence
- Do **not** invent user numbers, energy savings, or institutional adoption

Do Not Use Unsupported Statistics

- Verify all external statistics before using them
- If a number is not confirmed,